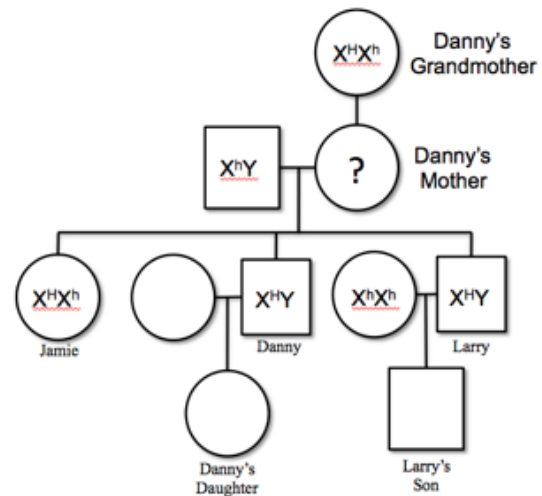


Take a deep breath. Relax. Choose the *best* answer. Clearly *circle* the letter for your answer. Vague or unclear marks will be counted as incorrect. Each question is worth 2 points.

1. One type of webbed toes is a Y-linked trait. (Y-linked traits are found only on the Y chromosome.) If a male with webbed toes marries a female that does not have the trait, their children might have which of the following phenotypes?
 - A. Normal females, web-toed females, normal males, and web-toed males.
 - B. Normal females, normal males or web-toed males.
 - C. Normal females, web-toed females, and web-toed males.
 - D. Normal females, web-toed females, and normal males.
 - E. Normal females and web-toed males.
 2. Fragile X syndrome is a sex-linked dominant trait that causes a range of developmental problems including learning disabilities and cognitive impairment. Usually, the disease is worse in males than females. Mary has the trait but her husband Mark does not. Their daughter, Michelle, does not have the trait. What is the probability that their next two children will both have the trait?
 - A. 75%
 - B. 50%
 - C. 100%
 - D. 25%
 - E. 12.5%
 3. After using the polymerase chain reaction to amplify the amelogenin gene from a blood sample, you find two distinct peaks (or bands) in the gel. What can you conclude?
 - A. The blood sample came from a male.
 - B. The blood sample came from the victim.
 - C. The blood sample came from a female.
 - D. The blood sample did not come from a human.
 - E. The blood sample came from the suspect.
 4. The sugary gene contributes to the sweetness of corn. A corn plant that is Ss for the sugary gene is crossed with a plant that is ss. What is the expected genotypic ratio of offspring?
 - A. 1 Ss : 1 ss
 - B. 1 SS : 1 Ss : 1 ss
 - C. 1 SS : 1 ss
 - D. 1 SS : 2 Ss : 1 ss
 - E. 1 SS : 1 Ss
-

5. Danny Ramos Gomez, his brother Larry, and his sister Jamie, all have hypertrichosis. Their disease is a sex-linked, dominant trait with alleles H and h . Based on his family pedigree, which of the following will be true for Danny's daughter and Larry's son?

- A. Danny's daughter and Larry's son both will have hypertrichosis.
- B. Danny's daughter and Larry's son both will not have hypertrichosis.
- C. Danny's daughter will have hypertrichosis. Larry's son will not have hypertrichosis.
- D. Danny's daughter may or many not have hypertrichosis, depending on the genotype of his wife. Larry's son will have hypertrichosis.
- E. Danny's daughter may or many not have hypertrichosis, depending on the genotype of his wife. Larry's son will not have hypertrichosis.



6. Assume you cross a black-feathered chicken with a white-feathered chicken, then cross two offspring from the F_1 generation. In the F_2 generation, $1/4$ of the chickens have black feathers, $1/4$ of the chickens have white feathers, and $1/2$ of the chickens are speckled. When you look closely, you notice that the speckles are caused by equal numbers of white and black feathers on the body. This type of inheritance is called
- A. codominance.
 - B. environmental effects.
 - C. incomplete dominance.
 - D. continuous dominance.
 - E. complete dominance.
7. The two strands of DNA are held together by
- A. three bonds between A and G, and two bonds between C and T.
 - B. three bonds between C and G, and three bonds between A and T.
 - C. three bonds between A and T, and two bonds between C and G.
 - D. three bonds between C and G, and two bonds between A and T.
 - E. two bonds between C and T, and two bonds between A and G.
8. The generation that results from crosses between true-breeding parents is called
- A. the P_1 generation
 - B. the true-breeding generation.
 - C. the F_1 generation.
 - D. the inbred generation.
 - E. the F_2 generation.

9. Compare the DNA sequence of the parent cell to the sequence of a daughter cell. Which one type of mutation best explains this difference?

Parent: 5' – CCGTTGAAGTACATTGAAGCTTT – 3'

Daughter: 5' – CCGTTGAAGTATATTGAAGCTTT – 3'

- A. point mutation
 - B. gene duplication
 - C. jumping gene
 - D. transposable element
 - E. frameshift mutation
10. After a plant breeder crossed two pink carnations, 24% of the offspring had red flowers, 54% had pink flowers, and 22% had white flowers. According to these results, both alleles are most likely
- A. incompletely dominant.
 - B. dominant.
 - C. recessive.
 - D. sex-linked.
 - E. codominant.

11. Which sequence below would make a good PCR primer for the following target gene?

Target gene: 3' – TCGGAATCTCTGGTCAATGCC – 5'

- A. 5' – GATTCCG – 3'
 - B. 3' – TCGGAAT – 5'
 - C. 5' – AGCCTTA – 3'
 - D. 3' – AGCCTTA – 5'
 - E. 3' – GATTCCG – 5'
 - F. 5' – TCGGAAT – 3'
12. Which of the following is not a nucleotide?
- A. Guanine
 - B. Urocine
 - C. Adenine
 - D. Thymine
 - E. Cytosine
13. The _____ strands of DNA form a _____ held together by _____ bonds.
- A. two, double helix, hydrogen.
 - B. one, single helix, hydrogen.
 - C. two, double helix, covalent.
 - D. one, double helix, nitrogen.
 - E. two, single helix, carbon.
-

14. A change in the nucleotide sequence of one or more genes is called
- A. mutation.
 - B. recombination.
 - C. phenotype.
 - D. deletion (like the amelogenin gene).
 - E. genotype.
15. The kernels on ears of corn come in a variety of colors and textures. The *P* gene controls color. The dominant *P* allele produces purple kernels. The recessive *p* allele produces yellow kernels. The *S* gene controls texture. The dominant *S* allele produces smooth kernels. The recessive *s* allele produces wrinkled kernels. If you cross a *ppss* individual with a *PpSs* individual, the expected genotype ratio for corn kernels in the next generation is
- A. 50% *PpSs*: 25% *Ppss*: 25% *ppss*
 - B. 25% *ppSs*: 50% *Ppss*: 25% *ppss*
 - C. 25% *PpSs*: 25% *Ppss*: 25% *ppSs*: 25% *ppss*
 - D. 25% *PpSs*: 50% *ppSs*: 25% *Ppss*
 - E. 9 *PpSs* : 3 *Ppss* : 3 *ppSs* : 1 *ppss*
16. The one allele that determines the phenotype of a heterozygote is
- A. the recessive gene.
 - B. the dominant allele.
 - C. the allele on the dominant gene.
 - D. the dominant gene.
 - E. the recessive allele.
17. DNA is a long, chainlike molecule made up of subunits called
- A. genes.
 - B. proteins.
 - C. base pairs.
 - D. nucleotides.
 - E. helices (plural of helix).
18. In a cross between two heterozygotes, the Punnett square shows the three possible _____ and two possible _____ that may appear in the offspring.
- A. allele; genotypes
 - B. genotypes; alleles
 - C. phenotypes; genes
 - D. phenotypes; genotypes
 - E. genotypes; phenotypes
-

19. Dogs have a hereditary form of deafness caused by the recessive d allele (not sex-linked). Betty has a dog that can hear but she does not know whether the dog is DD or Dd . She breeds her dog to one that is deaf. If Betty's dog is homozygous, what is the probability of producing deaf puppies?
- A. 75%.
 - B. 100%.
 - C. 25%.
 - D. 50%.
 - E. 0%.
20. Mutations which occur in
- A. germ-line and somatic tissues cannot be passed on to future generations.
 - B. germ-line and somatic tissues can be passed on to future generations.
 - C. germ-line tissues can be passed on to future generations.
 - D. germ-line tissues cannot be passed on to future generations.
 - E. somatic tissues can be passed on to future generations.
21. In cats, the agouti gene (A/a) determines the color of each hair. The dominant A allele produces orange hairs with black tips. The recessive a allele produces black hairs. The tabby gene (M/m) produces one of two tabby patterns. The dominant M allele produces the striped tabby pattern. The recessive m allele produces the classic spiral tabby pattern. If you cross a cat that is $AaMm$ with a cat that is $AaMm$, the expected phenotype ratio is
- A. 1 striped tabby with orange hair : 3 striped tabby with black hair : 3 classic tabby with orange hair : 1 classic tabby with black hair.
 - B. 9 striped tabby with orange hair : 3 striped tabby with black hair : 3 classic tabby with orange hair : 1 classic tabby with black hair.
 - C. 3 striped tabby with orange hair : 3 striped tabby with black hair : 1 classic tabby with orange hair : 1 classic tabby with black hair.
 - D. 1 striped tabby with orange hair : 2 striped tabby with black hair : 2 classic tabby with orange hair : 1 classic tabby with black hair.
 - E. 1 striped tabby with orange hair : 1 striped tabby with black hair : 1 classic tabby with orange hair : 1 classic tabby with black hair.
22. Based on Gregor Mendel's work, we know that traits are
- A. determined by independent assortment.
 - B. determined by random segregation.
 - C. inherited through the non-random passing of genes from parents to offspring.
 - D. inherited through the random passing of alleles from parents to offspring.
 - E. determined by a random combination of dominant and recessive alleles.
-

23. Assume one individual is Aa for the first trait, and tt for the second trait. The second individual is aa for the first trait and Tt for the second trait. The two traits are independent. What is the probability of obtaining an offspring that is $AaTt$? (Hint: Remember your rules for probability. You do not have to solve this using a dihybrid cross.)
- A. 0.625
 - B. 0.25
 - C. 0.75
 - D. 0.125
 - E. 0.5

24. In genetics, the principles of probability can be used to
- A. state with certainty the traits of the first born child.
 - B. decide which organisms are best to use in genetic crosses.
 - C. predict the traits of the offspring produced by genetic crosses.
 - D. predict the traits of the parents used in genetic crosses.
 - E. determine the actual outcomes of genetic crosses.

25. A crime suspect has alleles 9 and 12 for the CSF1PO short tandem repeat. What is the probability that someone else at random would have these same two alleles?

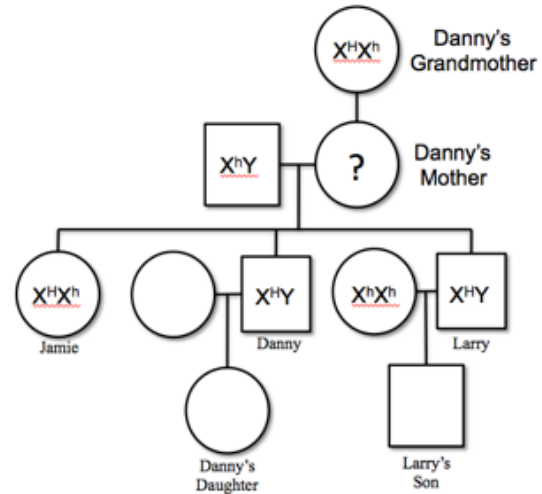
- A. 1/120.
- B. 1/60.
- C. 1/403.
- D. 1/40.
- E. 1/3.
- F. 1/43.

allele	frequency
9	1/40
10	1/5
11	1/3
12	1/3
13	1/10

26. DNA strands are
- A. antiparallel. 5' matches with 3'.
 - B. antiparallel. 5' matches with 5'.
 - C. parallel. 3' matches with 3'.
 - D. complementary. 5' matches with 3'.
 - E. complementary. 5' matches with 5'.
 - F. parallel. 5' matches with 5'.
27. A type of mutation that occurs when a piece of parasitic DNA inserts a copy of itself on another chromosome is called:
- A. a point mutation
 - B. a frameshift mutation
 - C. gene duplication
 - D. an intron
 - E. a jumping gene

28. Danny Ramos Gomez, his brother Larry, and his sister Jaime, all have hypertrichosis. Their disease is a sex-linked, dominant trait with alleles H and h . Based on his family pedigree, what is the genotype of Danny's mother?

- A. All three genotypes are possible for Danny's mother.
- B. $X^H X^H$ or $X^H X^h$
- C. $X^h X^h$
- D. $X^H X^h$
- E. $X^H X^H$



29. Duchene's muscular dystrophy (DMD) is a sex-linked recessive trait. Neither Betty nor her husband Bob has the trait. Their son, Buddy, was born with DMD. What is the chance that their next son will have DMD?

- A. 25%.
- B. 75%.
- C. 100%.
- D. 50%.
- E. 0%.

30. The figure on the next page shows the DNA fingerprint for five suspects, plus the evidence from the crime scene. Based on the DNA fingerprint, which suspect is the best match to the evidence?

- A. Suspect 1
- B. Suspect 2
- C. Suspect 3
- D. Suspect 4
- E. Suspect 5

